Online Appendix

Sovereign Risk Spillovers across Countries and Sectors

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This appendix first establishes the repayment-floor capacity results. It then explains the design and construction of the public-data exposure network, separating source observations, bilateral allocations and accounting reconciliation from the assumptions governing distress. A final section documents the Eurosystem purchase experiment. The baseline is the end-2024 residence network with public OCR headroom, full accounting equity and bank $\lambda = .90$.

1 Capacity of the repayment-floor guarantee

This argument applies to the bank-clearing system with fixed lending positions. Fix a sovereign state s. Let $\mathcal{B}$ denote the bank blocks and x_{ij} creditor bank i's claim on debtor bank j. All inequalities between repayment vectors are componentwise. For each bank i, normalize contractual repayment to one and assume $D_i > 0$, $0 < \lambda_i^B \leq 1$, and nonnegative bilateral exposures and receipts from nonbank assets. The guarantee is financed outside this block, so its payments do not reduce any bank's other resources. Write $L_j^{BB} = \sum_{i \in \mathcal{B}} x_{ij}$ for debtor j's obligations to bank creditors and $[z]_+ = \max\{z, 0\}$, applied componentwise to vectors. Retain the preferred repayment floors $\bar{R}_j(s)$, the greatest original equilibrium, and the compensation benchmark B_{post} defined in the main text.

1.1 A sufficient capacity

The repayment map without intervention For a candidate repayment vector $\mathbf{R} = (R_i)_{i \in \mathcal{B}}$, write bank i's asset receipts without support as

$$A_i^0(\mathbf{R}; s) = a_i(s) + \sum_{j \in \mathcal{B}} x_{ij} R_j, \qquad a_i(s) \geq 0.$$

The nonbank component $a_i(s)$ is independent of bank repayments. Let $\mathcal{K} = [0,1]^{|\mathcal{B}|}$ and fix a resource cutoff $0 < T_i \leq D_i/\lambda_i^B$. The ordinary repayment from resources $A \geq 0$ is the function $\phi_i(A)$ defined by

$$\phi_i(A) = \begin{cases} \lambda_i^B A / D_i, & A < T_i, \\ 1, & A \geq T_i. \end{cases} \qquad F_{s,i}(\mathbf{R}) = \phi_i\big(A_i^0(\mathbf{R}; s)\big).$$

Here E_i denotes accounting equity and $H_i \leq E_i$ prudential headroom. The model's cutoff is $T_i = \min\{D_i + E_i - H_i, D_i/\lambda_i^B\}$; $H_i = E_i$ gives the accounting-insolvency trigger. The upper bound on T_i ensures that default-branch repayments do not exceed the contractual rate. Banking equilibria without intervention are precisely the fixed points of F_s on $\mathcal{K}$. The map is nondecreasing and maps $\mathcal{K}$ into itself. Write $\text{Fix}(F_s)$ for its fixed-point set. By Tarski (1955), it has a least fixed point $\underline{\mathbf{R}}(s)$ and a greatest fixed point. No continuity assumption is needed for the budget argument below.

A comparison with an original equilibrium Consider any equilibrium $\mathbf{R}$ under nonnegative transfers $t_i(\mathbf{R}; s)$ to creditor banks. Its repayments satisfy

$$R_i = \phi_i\big(A_i^0(\mathbf{R}; s) + t_i(\mathbf{R}; s)\big) \geq F_{s,i}(\mathbf{R}).$$

For every $\mathbf{Z} \in [0, \mathbf{R}]$, monotonicity gives $0 \leq F_s(\mathbf{Z}) \leq F_s(\mathbf{R}) \leq \mathbf{R}$. Thus F_s maps the complete lattice $[0, \mathbf{R}]$ into itself. Tarski's theorem supplies an original fixed point in this interval, so the least

original fixed point satisfies $\underline{\mathbf{R}}(s) \leq \mathbf{R}$. This order comparison requires no adjustment dynamics and holds even if the policy repayment map is not monotone.

Let the total claims on the repayment-floor guarantee at any repayment vector be

$$H_s(\mathbf{R}) = \sum_{j \in \mathcal{B}} L_j^{BB} [\bar{R}_j(s) - R_j]_+.$$

Equivalently, the compensation benchmarks in the main text are

$$C_{B,\max}(s) = \max_{\mathbf{Q} \in \mathrm{Fix}(F_s)} H_s(\mathbf{Q}) = H_s\big(\underline{\mathbf{R}}(s)\big), \qquad B_{\mathrm{post}} = \max_s C_{B,\max}(s).$$

Since H_s is nonincreasing in each repayment, the comparison gives

$$H_s(\mathbf{R}) \leq H_s\big(\underline{\mathbf{R}}(s)\big) = C_{B,\max}(s) \leq B_{\mathrm{post}}. \tag{1}$$

Thus the full claims at any equilibrium supported by nonnegative transfers are bounded by the compensation bill at the least original equilibrium. Intermediate original equilibria are not needed to compute this bound.

A complete capped payment rule For a budget $B \geq 0$, define the proportional payment fraction

$$\theta_B(\mathbf{R}; s) = \begin{cases} 1, & H_s(\mathbf{R}) \leq B, \\ B / H_s(\mathbf{R}), & H_s(\mathbf{R}) > B. \end{cases}$$

The second case has a strictly positive denominator, including when $B = 0$. Bank creditors of debtor j receive the effective rate

$$\widetilde{R}_j^{(B)} = R_j + \theta_B(\mathbf{R}; s)[\bar{R}_j(s) - R_j]_+.$$

Equivalently, bank i receives the transfer

$$t_i^{(B)}(\mathbf{R}; s) = \theta_B(\mathbf{R}; s) \sum_{j \in \mathcal{B}} x_{ij} [\bar{R}_j(s) - R_j]_+.$$

A capped equilibrium is a vector $\mathbf{R} \in \mathcal{K}$ satisfying

$$R_i = \phi_i\big(A_i^0(\mathbf{R}; s) + t_i^{(B)}(\mathbf{R}; s)\big) \quad \text{for every bank } i.$$

Thus creditors anticipate any rationing when valuing their interbank assets. Total expenditure is $\min\{H_s(\mathbf{R}), B\}$. The uncapped guarantee sets the payment fraction to one at every repayment vector.

Proposition 1 (Sufficient budget)**.** *Under the assumptions above, for every $B \geq B_{\mathrm{post}}$, the capped and uncapped repayment-floor guarantees have the same banking equilibria in every sovereign state. Their payments also coincide at those equilibria.*

Proof. At a capped equilibrium, nonnegative transfers imply (1). Claims therefore fit within the budget, so $\theta_B(\mathbf{R}; s) = 1$. The repayment equations coincide with those of the uncapped guarantee. Conversely, transfers at any uncapped equilibrium are nonnegative, so the same bound implies $H_s(\mathbf{R}) \leq B$. The capped rule then pays all claims in full and supports that equilibrium with identical payments. This proves equality of the equilibrium sets. □

The argument extends to any nonnegative allocation rule that pays all claims in full whenever $H_s(\mathbf{R}) \leq B$; proportional rationing merely specifies the policy when claims exceed the cap. The cap may ration claims away from equilibrium. The proposition establishes that it never rations claims at an equilibrium when $B \geq B_{\mathrm{post}}$. Combined with the main-paper implementation proposition, this capacity bound gives unique implementation of the greatest-equilibrium target. Financing that withdraws resources from banks is outside the proposition.

1.2 The critical capacity under proportional rationing

Fix s and write $\mathbf{Q} = \bar{\mathbf{R}}(s)$, $\mathbf{h}(\mathbf{R}) = (\mathbf{Q} - \mathbf{R})_+$ and $H(\mathbf{R}) = H_s(\mathbf{R})$. Let

$$G_B(\mathbf{R}) = F_s(\mathbf{R} + \theta_B(\mathbf{R}; s)\mathbf{h}(\mathbf{R})), \qquad G^\theta(\mathbf{R}) = F_s(\mathbf{R} + \theta\mathbf{h}(\mathbf{R})).$$

Define $\Gamma_s = \{(\theta, \mathbf{R}) : 0 \le \theta < 1,\ \mathbf{R} \in \mathcal{K},\ \mathbf{R} = G^\theta(\mathbf{R}),\ H(\mathbf{R}) > 0\}$ and

$$B_s^\dagger = \sup_{(\theta,\mathbf{R})\in\Gamma_s} \theta H(\mathbf{R}), \qquad B_s^\dagger = 0 \text{ if } \Gamma_s = \varnothing.$$

The condition $H(\mathbf{R}) > 0$ excludes the target and zero-claim vectors, for which the policy sets $\theta_B = 1$.

Proposition 2 (Critical capacity)**.** *The target* $\mathbf{Q}$ *is the greatest fixed point of* G_B *for every* $B \ge 0$*. A nonpreferred capped equilibrium exists at capacity* B *exactly when* $B = \theta H(\mathbf{R})$ *for some* $(\theta, \mathbf{R}) \in \Gamma_s$*. Capacities admitting such an equilibrium form a downward-closed set with supremum* $B_s^\dagger$*. Thus all* $B > B_s^\dagger$ *implement* $\mathbf{Q}$ *uniquely, and all* $0 \le B < B_s^\dagger$ *admit another equilibrium. Equality implements uniquely precisely when no pair in* Γ_s *attains the supremum.*

Proof. For fixed $\theta \in [0, 1]$, each effective receipt $R_j + \theta[Q_j - R_j]_+$ is nondecreasing in both R_j and θ. Since H is nonincreasing in $\mathbf{R}$, the paid fraction θ_B is nondecreasing in $\mathbf{R}$ and in B. Thus G_B is a nondecreasing self-map of $\mathcal{K}$ and increases with B. In particular, $H(\mathbf{Q}) = 0$ implies $G_B(\mathbf{Q}) = \mathbf{Q}$.

The fully honoured map $G^1(\mathbf{R}) = F_s(\mathbf{R} \vee \mathbf{Q})$ has the unique fixed point $\mathbf{Q}$ by the main-paper implementation proposition. Any fixed point $\mathbf{R}$ of G_B or G^θ satisfies $\mathbf{R} \le G^1(\mathbf{R})$. Monotonicity therefore makes G^1 a self-map of the interval $[\mathbf{R}, \mathbf{1}]$. Tarski supplies a fixed point in that interval; it must be $\mathbf{Q}$. Hence $\mathbf{R} \le \mathbf{Q}$, and $\mathbf{Q}$ is the greatest capped fixed point.

If a capped fixed point differs from $\mathbf{Q}$, it must have $H(\mathbf{R}) > B$, since otherwise it solves the fully honoured equations. Setting $\theta = B/H(\mathbf{R}) < 1$ gives a pair in Γ_s with $B = \theta H(\mathbf{R})$. Conversely, for any pair in Γ_s, $H(\mathbf{R}) > 0$ and $\theta < 1$ imply that capacity $B = \theta H(\mathbf{R})$ pays exactly fraction θ. It therefore supports the same nonpreferred fixed point. This proves the correspondence.

To establish the threshold, suppose B_2 admits a nonpreferred fixed point $\mathbf{R}_2$ and $0 \le B_1 \le B_2$. Then $G_{B_1}(\mathbf{R}_2) \le G_{B_2}(\mathbf{R}_2) = \mathbf{R}_2$, so G_{B_1} maps $[\mathbf{0}, \mathbf{R}_2]$ into itself. Tarski gives a fixed point $\mathbf{R}_1 \le \mathbf{R}_2 \le \mathbf{Q}$, with $\mathbf{R}_1 \ne \mathbf{Q}$. The nonimplementing capacities are therefore downward closed. The correspondence identifies their supremum with $B_s^\dagger$ and establishes the endpoint criterion. If $\Gamma_s = \varnothing$, every capacity implements uniquely, including $B = 0$. □

For $(\theta, \mathbf{R}) \in \Gamma_s$, $F_s(\mathbf{R}) \le \mathbf{R}$, so the earlier interval argument gives $\underline{\mathbf{R}}(s) \le \mathbf{R}$. Hence $\theta H_s(\mathbf{R}) \le H_s(\underline{\mathbf{R}}(s)) = C_{B,\max}(s)$; taking suprema and maxima proves $B^\dagger \le B_{\text{post}}$. For common capacity $B = B^\dagger$, only states attaining that maximum need the endpoint test. Neither the sufficient-bound nor critical-capacity comparison requires continuity or Eisenberg–Noe regularity. Nonattainment is familiar in unique-implementation problems (Segal, 2003). Even in a scalar version of the abstract isotone model, an infimum need not be an attained minimum: for $F(R) = R$ on $[0, 1]$, $Q = 1$ and $H(R) = 1 - R$, any $B > 0$ implements uniquely but $B = 0$ does not.

1.3 Numerical evaluation by default sets

Let $\mathbf{B} = (x_{ij})$ be the bank claim matrix, $w_j = \sum_i x_{ij} = L_j^{BB} > 0$ and $\mathbf{a} = \mathbf{a}(s)$. Bold symbols denote matrices and vectors; their subscripts select the indicated blocks. Let $\mathbf{w} = (w_j)$, let $\mathbf{T} = (T_i)$ collect the resource cutoffs, and let $\mathbf{I}$ and $\mathbf{1}$ denote an identity matrix and a vector of ones of matching dimensions. For a candidate default set J and nondefaulting complement $S = \mathcal{B} \setminus J$, $\mathbf{Q}_S = \mathbf{1}$ is necessary since all policy roots lie below $\mathbf{Q}$. With $\mathbf{\Gamma}_J = \operatorname{diag}(\lambda_i^B / D_i)_{i\in J}$, the fixed-fraction equations give

$$\begin{aligned} \mathbf{R}_S(\theta) &= \mathbf{1}, \\ \mathbf{R}_J(\theta) &= [\mathbf{I} - (1-\theta)\mathbf{\Gamma}_J\mathbf{B}_{JJ}]^{-1}\mathbf{\Gamma}_J[\mathbf{a}_J + \mathbf{B}_{JS}\mathbf{1} + \theta\mathbf{B}_{JJ}\mathbf{Q}_J]. \end{aligned} \tag{2}$$

Retain candidates only if $0 \le \mathbf{R} \le \mathbf{Q}$, $\mathbf{R} \ne \mathbf{Q}$ and, for $\mathbf{V} = \mathbf{a} + \mathbf{B}[(1-\theta)\mathbf{R} + \theta\mathbf{Q}]$,

$$\mathbf{V}_J < \mathbf{T}_J, \qquad \mathbf{V}_S \ge \mathbf{T}_S.$$

Their associated capacity is $B_J(\theta) = \theta \mathbf{w}_J^\top [\mathbf{Q}_J - \mathbf{R}_J(\theta)]$. Maximizing its supremum over all admissible default regimes computes $B_s^\dagger$.

A sufficient condition for the inverse to exist is

$$\nu = \max_i \lambda_i^B w_i / D_i < 1.$$

Let $\mathbf{A} = \mathbf{\Gamma}_J \mathbf{B}_{JJ}$. For every $j \in J$,

$$(\mathbf{w}_J^\top \mathbf{A})_j = \sum_{i \in J} \frac{\lambda_i^B w_i}{D_i} B_{ij} \le \nu \sum_{i \in J} B_{ij} \le \nu w_j.$$

Thus $\mathrm{spr}(\mathbf{A}) \le \nu < 1$. For $\mathbf{M} = \mathbf{I} - (1-\theta)\mathbf{A}$ and $0 \le \theta \le 1$, the inverse is the nonnegative Neumann series $\mathbf{M}^{-1} = \sum_{k \ge 0} (1-\theta)^k \mathbf{A}^k$. The determinant of $\mathbf{M}$ cannot vanish and equals one at $\theta = 1$, so $\det \mathbf{M} > 0$ throughout. In the data $\nu \simeq .15$.

Within an admissible regime let $\mathbf{g} = \mathbf{Q}_J - \mathbf{R}_J(\theta) \ge 0$, with $\mathbf{g} \ne 0$. Equation (2) gives the constant vector

$$\mathbf{M}\mathbf{g} = \mathbf{Q}_J - \mathbf{\Gamma}_J [\mathbf{a}_J + \mathbf{B}_{JS}\mathbf{1} + \mathbf{B}_{JJ}\mathbf{Q}_J].$$

Differentiation yields $\mathbf{g}' = -\mathbf{M}^{-1}\mathbf{A}\mathbf{g}$. Since $\mathbf{A}$ commutes with $\mathbf{M}^{-1}$,

$$B_J'(\theta) = \mathbf{w}_J^\top (\mathbf{I} - \mathbf{A}) \mathbf{M}^{-1} \mathbf{g} \ge (1-\nu) \mathbf{w}_J^\top \mathbf{M}^{-1} \mathbf{g} > 0.$$

The strict inequality uses positive weights and $\mathbf{M}^{-1}\mathbf{g} \ge \mathbf{g} \ne 0$. Thus the capacity supremum on an admissible interval is its right-hand limit. Endpoint inclusion must still be checked against the strict default inequalities.

The numerical calculation therefore needs only the admissible intervals and their right endpoints; no search for interior maxima is required. For m defaulting banks, $\det \mathbf{M}$ has degree at most m in θ. Since $\mathbf{M}\mathbf{g}$ is constant, the numerator of $\mathbf{g}$ has degree at most $m-1$. Clearing denominators in the default and repayment constraints gives boundary polynomials of degree at most m, hence at most four in our application.

The R implementation enumerates the possible default sets, numerically solves these boundary polynomials, and evaluates capacity at the right-hand limit of each admissible interval. It checks isolated boundary candidates and whether each endpoint satisfies the strict default inequalities. At a specified capacity, the equation $B_J(\theta) = B$ is also polynomial after clearing denominators and identifies adverse equilibria within each branch. Direct linear-system calculations and monotone iterations provide independent numerical checks away from cutoffs. An upward iteration at a discontinuity can approach a nonfixed left limit, so iteration alone cannot determine endpoint inclusion. The reported thresholds are numerical approximations, with equation residuals and boundary tolerances checked separately; the calculation is not a symbolic closed-form solution or a grid over θ.

1.4 Quantitative application and positioning

The common critical capacity is approximately EUR 60.6 billion, or 0.29% of aggregate clearing debt, compared with B_{post} = EUR 126.3 billion. The maximum is reached in the all-sovereign-default state. The last adverse branch there has limiting ordinary repayments

$$(R_{\text{DEU}}, R_{\text{FRA}}, R_{\text{ITA}}, R_{\text{ESP}}) \simeq (100\%, 95.1\%, 92.8\%, 92.9\%).$$

Its full floor claims approach EUR 67.2 billion and θ approaches .90. French assets reach T_{FRA} at that endpoint. Full repayment then applies, so the adverse default candidate is excluded. The same French-cutoff exit determines the implementing threshold in each of the other seven multiple states. Thus all positive state thresholds are attained minima under the maintained tie convention. The other eight states already implement uniquely at zero capacity. The main-paper table reports the state thresholds, rounded to one decimal place.

The target retains fundamental Italian and Spanish bank defaults in the all-sovereign-default state and entails zero transfers. The all-four-bank-default branch ends at about EUR 2.9 billion. This explains why removing the most adverse root need not remove all multiplicity. Numerical evaluation checks all admissible default regimes and their boundary conditions.

The coordination mechanism is established (Jackson and Pernoud, 2021); minimum transfers and worst-equilibrium implementation are central to Jackson and Pernoud (2024). Our result characterizes capacity for a specified proportional shortfall rule. Its repayment target can retain fundamental bank defaults. The distinctive policy feature is that creditors anticipate endogenous rationing of repayment-contingent support. The fixed-point comparisons are standard (Tarski, 1955), and commitment capacity already matters in sovereign backstop models (Roch and Uhlig, 2018). The characterization and finite evaluation quantify the commitment capacity required by this repayment-contingent policy.

These statements hold with fixed portfolios, fixed nonbank receipts and external financing that does not remove bank resources. The sufficient-capacity argument also applies to other fixed floors in $\mathcal{K}$ if the compensation bound is recomputed for those floors. Unique implementation and the critical-capacity result use the greatest original equilibrium as target. Endogenous nonbank repayment outside the par certificate, endogenous portfolios and resource-withdrawing financing require separate analysis.

2 Design of the empirical network

2.1 What the reconstruction needs to measure

The empirical object is a matrix of debt claims linking sectors and countries at the end of 2024. Its purpose is to identify who loses receipts when a sovereign or a bank repays less, and how those losses affect other banks' ability to pay. This requires consistent creditor and debtor positions, sovereign holdings by banks, and a reconciliation with bank liabilities and equity. It does not require a microeconomic model of how each portfolio was chosen or a complete network of every financial instrument.

No public source supplies this matrix directly. Banking statistics provide strong controls on resident bank balance sheets but incomplete bilateral detail. Financial accounts cover other sectors but often aggregate foreign counterparties. Portfolio surveys provide cross-border securities positions with uneven sector coverage. The construction therefore combines these sources in a hierarchy: published instrument totals anchor the accounts, available bilateral observations guide their allocation, and missing detail is filled within the applicable controls. The exposure register distinguishes these cases and retains the adjustments.

2.2 The main choices, especially for banks

A common residence perimeter. Bank nodes represent offices located in each country, including foreign-owned offices and excluding the foreign affiliates of domestic groups. This matches the residence basis of the financial accounts and public securities positions. Bank for International Settlements (BIS) consolidated claims instead describe worldwide banking groups; combining them directly with resident funding and equity would mix institutions and foreign affiliates across the

two sides of the balance sheet. We consequently use European Central Bank (ECB) resident-bank statistics as the bank controls. This preserves a common statistical perimeter without claiming that every bilateral position is observed.

Bank assets and funding from their own source controls. Public bank loan data distinguish firms and households, subject to the reporting-sector adjustments described below. Household bank funding comes from depositor-sector observations and securities holdings. Equity follows the published capital accounts, with a matched asset adjustment for provisions. These stocks are measured separately: missing funding classifications remain explicit, and the exposure matrix is subsequently reconciled to the complete balance sheet. This keeps creditor incidence and bank loss absorption tied to their relevant public sources.

Sovereign exposures anchored in bank portfolios. Aggregate bank government-security holdings are controlled by ECB balance-sheet statistics. Domestic securities holdings at face value are preferred where public; Italy requires an allocation within its bank portfolio control. Foreign holdings use residence-based portfolio information and bilateral securities envelopes. These allocations must respect both bank instrument controls and issuer totals. Public government debt-security stocks anchor the sovereign columns, while valuation differences and adjustments remain recorded. Matching the total debt stock does not make every bilateral bank holding an independently observed face amount.

Remove identified intragroup claims and retain other domestic links. Related-office loan margins permit removal of identified intragroup positions, within the reconstructed gross exposures and with matched bank balance-sheet adjustments. The remaining domestic claims are retained as gross obligations among distinct banks. The result is neither a worldwide consolidated group network nor a fully consolidated national banking sector. Public data do not identify every intragroup security position or foreign-office link. The model's common repayment rate within each country is a separate aggregation assumption, whose sensitivity is reported.

Incomplete links remain distinguishable from observations. Missing bilateral detail is allocated using published margins, available upper bounds and stated priors. Unidentified holders and issuers have explicit reporting accounts. Loan assets and their funding mirrors are reconciled rather than added as separate claims; bank securities drawn from different sources are likewise not counted twice. Assets outside the selected debt instruments and unresolved coverage or valuation differences enter the bank balance-sheet reconciliation. They are not treated as independently measured bilateral exposures.

The resulting database is a controlled reconstruction, not an observation of the complete underlying network. Its strength is a consistent public-data account of the positions relevant to repayment, with imputations and limits visible. Its principal uncertainties concern bilateral allocations, valuation and the treatment of intragroup and gross domestic claims. Prudential headroom, private-credit crisis losses and default costs enter only in the subsequent model mapping; they do not determine the matrix entries.

2.3 How to read the construction

Section 3 gives the operational sequence: define coverage and source controls; construct bank loans and funding; reconcile interbank claims and related offices; allocate securities holdings; and check issuer totals and unknown counterparties. Section 4 then reconciles these selected claims with full bank balance sheets and explains the crisis specification. This separates what is measured or imputed from how a given loss changes repayment.

3 Construction of the cross-country sectoral exposure database

This section implements the design described in Section 2. The construction starts from published sector and instrument controls, allocates the detail needed for bilateral transmission, and records every reconciliation. Amounts are EUR billions unless stated otherwise. The final matrix, intermediate controls and long exposure register accompany the replication files.

3.1 Target object, coverage and reference date

Let $\mathcal{C} = \{\text{DEU}, \text{FRA}, \text{ITA}, \text{ESP}\}$. Each country has five sectors: non-financial corporations (S11), banks (S122), nonbank financial institutions (S12nbfi), general government (S13), and households including non-profit institutions serving households (NPISHs) where the source combines them (S14). Nonbank financial institutions comprise S123–S129, including money market funds (MMFs). Banks exclude MMFs and central banks.

The reporting matrix $\mathbf{W}$ has creditor rows and debtor columns. It contains these twenty country–sector nodes, an aggregate central-bank node EA:S121, the rest of the world ROW:S2, and two classification accounts, UNALLOC:ISSUER and UNALLOC:HOLDER. It is therefore a 24×24 matrix. The classification accounts retain positions with an unidentified sector or holder; they are not additional economic sectors. Country information that survives in the source remains in the exposure register even when the matrix uses an unallocated account.

For model exposition and the full sector-network view, ROW:S2, UNALLOC:ISSUER and UNALLOC:HOLDER are grouped into a single passive node, Other counterparties. The resulting matrix has 22 accounts and retains the Eurosystem separately. Aggregation sums gross creditor and debtor positions without netting them. It preserves each identified bilateral claim, total bank assets and liabilities represented in the matrix, and the instrument-specific distinction between ordinary bank recovery and fixed settlements. The reporting matrix and exposure register retain the geographical and classification detail, since an unidentified domestic holder is not a foreign investor.

The retained instruments are bank loans, deposits and debt securities, and nonbank debt-security holdings and deposits with banks. Government loans remain in the separate balance-sheet reconciliation because the sovereign shock concerns debt securities. Equity, fund shares, derivatives and other assets are not reconstructed as a complete network of promised debt payments. Statistical remaining liabilities are reported separately by instrument in the register, including when they enter an unallocated matrix cell.

The database is a snapshot at 31 December 2024. Quarter-start dates such as `2024-10-01` label end-quarter positions. December monthly stocks and fourth-quarter stocks are alternatives, never additive observations. The annual Portfolio Investment Positions (PIP) observations used here refer to the same year end. No earlier quarterly position is inferred from the annual observation for this construction. ECB and IMF source snapshots were retrieved on 21 October 2026; Eurostat inputs retain the recorded extraction vintage. The replication files identify the source vintage for each input.

3.2 Source hierarchy and construction sequence

Table 1 summarizes the source hierarchy (ECB, 2026; Eurostat, 2026; IMF, 2026). BSI denotes Balance Sheet Items; SHSS denotes Securities Holdings Statistics; PIP was formerly the Coordinated Portfolio Investment Survey (CPIS). Source-table frequencies M and Q mean monthly and quarterly; financial-account codes F2, F3 and F4 denote currency and deposits, debt securities and loans. BIS consolidated claims and nationality-based domestic exposures do not enter the matrix. Worldwide group exposures cannot be spliced into resident-office balance sheets without a perimeter adjust-

ment. They may serve as benchmarks, with their consolidation and counterparty definitions stated separately.

Table 1: Sources and their roles in the residence-based construction

Source	Instruments and frequency	Retained role
ECB BSI	Loans, deposits, securities and balance-sheet totals; M/Q	Resident S122 controls; borrower and depositor sectors; bank and central-bank counterparties; related-office margins.
ECB SHSS	Debt securities; Q	Domestic bank and MMF holdings, with face and market values distinguished.
Eurostat financial accounts	Financial stocks; Q	Domestic nonbank holdings, government and central-bank portfolios, and domestic/nonresident issuer margins.
IMF PIP/CPIS	Portfolio debt securities; annual observation	Bilateral foreign portfolios and reported holder/issuer sector margins; constrained completion of missing sector detail.
Eurostat government debt	Consolidated nominal F3 stock; Q	Sovereign debt-security totals. Loans and other public-debt instruments are outside this anchor.
ECB ILM	Monetary-policy lending; M	Internal Liquidity Management statistics: refinancing diagnostic, not an addition to central-bank funding already recorded from BSI.

Construction proceeds in five steps: (i) establish resident bank instrument totals; (ii) allocate loan, deposit and securities counterparties; (iii) reconcile interbank positions and remove identified related-office loans; (iv) reconcile securities to issuer stocks while retaining unknown holders separately; and (v) reconcile selected claims to complete bank balance sheets. The source layers and all intermediate adjustments are retained. Aggregate sectors such as S1 discipline allocations but are never added to their component sectors as network nodes.

3.3 Bank loans and deposit funding

We first establish what banks lend and who funds them, using the same resident-bank controls on both sides. Three distinctions matter: bank versus money-market-fund reporting, the sector of private borrowers, and the sector of depositors. The adjustments below allocate those controls; they do not add a second layer of bank assets or funding.

Separate banks from money market funds. BSI reporting sector B identifies S122; sector A comprises monetary financial institutions (MFIs) excluding the European System of Central Banks (ESCB) and includes MMFs; sector F identifies MMFs. Where only sector-A bilateral detail is public, it is allocated within a matching sector-B instrument and counterparty class. For reporting country i, counterparty country j, instrument k and counterparty class g, let $T_i^{k,g,\mathrm{S122}}$ and $T_i^{k,g,\mathrm{MFI}}$ be the corresponding published portfolio totals, and $w_{ij}^{k,g,\mathrm{MFI}}$ the bilateral MFI position. Define

$$\rho_i^{k,g} = \frac{T_i^{k,g,\mathrm{S122}}}{T_i^{k,g,\mathrm{MFI}}}, \qquad \widehat{w}_{ij}^{k,g} = \rho_i^{k,g} w_{ij}^{k,g,\mathrm{MFI}}. \tag{3}$$

This is an allocation of an observed total, not an observation of the bilateral S122 position.

The register retains the MFI value, the allocation method and available bounds. For example, France's S122/MFI private-loan ratio is about 0.996: treating the MFI detail as a bank observation would retain some MMF lending.

For outside-euro-area loans and securities, published MMF instrument totals are subtracted directly when available.

Identify loan borrowers. Domestic and bilateral sample-country loan observations distinguish nonfinancial corporations (NFCs) and households/NPISHs. The financial borrower residual is calculated within the private-sector loan class, after excluding government loans. Missing bilateral government loans are allocated using the observed foreign-euro-area government share within the nonbank envelope. Unassigned euro-area borrowers are retained in UNALLOC:ISSUER. Household loans use the reported household borrower-sector observations.

Identify deposit funding. Household and NFC deposits are taken from their reported depositor-sector cells, with the bank/MMF adjustment in (3) where necessary. Foreign financial and government depositors use the available disjoint sector breakdowns. Missing domestic financial/government detail is recovered from euro-area sector totals less reported foreign-euro-area positions. Any unclassified foreign depositor budget is allocated within the observed domestic residual, with the adjustment and bounds recorded.

Domestic MMF depositors are separated from other MFIs and assigned to S12nbfi.

Outside-euro-area deposit funding is allocated between S122 and MMFs using their external-liability totals where the deposit split itself is unpublished. The remaining known external liabilities stay in an outside funding component. Unassigned euro-area deposits and other liabilities remain explicit. Household funding comprises measured deposits and separately reconciled bank securities.

3.4 Interbank loans: reconcile mirrors, then remove related offices

A loan from one bank is a liability of another. Asset-side reports and depositor-side mirrors provide complementary information on that same claim, but their public breakdowns need not agree cell by cell. We anchor the construction in loan asset totals and use funding mirrors and bilateral envelopes to allocate them. Identified related-office positions are removed only after the gross network has been formed.

The interbank loan construction covers all twenty countries belonging to the euro area at end-2024, before aggregating nonsample countries into ROW. Otherwise, exposures to the rest of the euro area would be forced into the four-country block. Central-bank counterparties are excluded from bank–bank loans and retained in EA:S121.

Let N_{ij} denote gross loans from banks in country i to banks in country j. The source provides bank loan asset margins, deposit-liability mirrors and, for many country pairs, broader bilateral MFI envelopes. Bank asset margins determine row totals. Foreign deposit mirrors are first rescaled to the same aggregate total. Available bilateral MFI values provide upper bounds; missing links receive a borrower-size prior based on foreign bank deposit margins. Domestic bank loan positions supply the diagonal, with the reporting-bank adjustment in (3).

The allocation is a nonnegative, bounded weighted projection. Let $\mathbf{P}$ be the prior matrix, $\mathbf{r}$ the vector of row margins, $\mathbf{c}$ the vector of column targets and $\mathbf{U}$ the matrix of cell caps. For allocation matrix $\mathbf{Z}$, a small positive weight floor ε, a vector of ones $\mathbf{1}$ and transpose denoted by $\top$, the basic problem is

$$\min_{\mathbf{Z}\geq 0} \sum_{ij} \frac{(Z_{ij}-P_{ij})^2}{\max(P_{ij},\varepsilon)} \quad \text{s.t.} \quad \mathbf{Z1}=\mathbf{r}, \quad \mathbf{Z}^\top\mathbf{1}=\mathbf{c}, \quad Z_{ij}\leq U_{ij}. \tag{4}$$

When mirror columns are incompatible with the row margins and caps, rows and caps remain hard constraints and column deviations enter a weighted least-squares penalty. The register reports the resulting asset/liability mirror discrepancies.[1]

BSI items A20G and L20G identify related-office loan/deposit margins. A related-office matrix $\mathbf{G}$ is fitted within the gross matrix, $0 \leq G_{ij} \leq N_{ij}$. Identified foreign asset-side margins are removed in full; funding mirrors are reconciled within the gross bounds. The domestic removal uses the common observed subset, bounded by the adjusted related-office asset margin, the funding margin and the gross domestic position. Net interbank loans are

$$L_{ij}^{BB} = N_{ij} - G_{ij}.$$

Missing auxiliary-country margins are estimated from available MFI/central-bank detail or median observed euro-area counterparty/related-office shares and flagged separately. Related securities and outside-euro-area related-office positions without a public split are not assumed absent.

Remaining domestic claims between independent banks are retained. Residence-based removal of identified related-office positions does not constitute worldwide group consolidation or full national-sector consolidation. The replication outputs also provide gross-related-office and zero-domestic-diagonal accounting variants. The model retains the adjusted gross domestic claims; a separate clearing run cancelling domestic bank claims and equal liabilities removes multiplicity at the reference parameters (Section 4).

3.5 Bank securities: portfolio controls and issuer allocation

The bank securities block follows a source hierarchy within observed portfolio totals. Sovereign holdings are treated first because they determine direct exposure to the model's initiating shock. Bank-issued and private securities are then allocated within their own controls. S122 securities totals distinguish MFI, government and private-sector issuers. Domestic SHSS face-value holdings are preferred for sovereign exposures where public (Germany, France and Spain). Italy's domestic sovereign holding is allocated from MFI geography within its S122 government-security control. Foreign government portfolios use residence-based CPIS sector priors where available and bilateral MFI nonbank-security positions elsewhere, subject to the latter envelopes. Private portfolios use the corresponding residual nonbank-security envelopes; NFC/NBFI detail uses domestic SHSS or CPIS/issuer-stock priors. Allocations to nonsample euro-area issuers enter ROW. An allocation that cannot be reconciled to an issuer remains unallocated while the bank's instrument total is preserved.

BSI and SHSS bank-security positions are not interchangeable. BSI excludes retained own securities from both holdings and issuance, whereas SHSS holdings can include them. Domestic bank-to-bank securities therefore use BSI-matched class and geography; SHSS is retained as a benchmark. Differences can also reflect valuation and allocation, so the full gap is not identified as own retention (ECB, 2024).

3.6 Domestic securities held by nonbank sectors

Domestic F3 holdings use Eurostat quarterly stock assets: frequency Q, `stk_flow=STK`, `finpos=ASS`, and `unit=MIO_EUR`. Transactions, other changes and liability observations are excluded. Household/NPISH sectors are mapped jointly where published that way. For financial subsectors, a reported S125–S127 aggregate takes precedence over its components; both are never added.

The domestic Eurostat holder slice contains no public S123 F3 observations. Domestic MMF holdings are therefore added from SHSS, with missing sector detail allocated within the domestic

[1] Amounts are in EUR bn. The implementation uses $\varepsilon = 10^{-5}$. With soft columns, the objective is 10^{-3} times the prior term in (4) plus $\sum_j [(\mathbf{Z}^\top \mathbf{1})_j - c_j]^2 / \max(c_j, 10^{-3})$. The positive denominator floor keeps the objective weights finite. All fitted margins and discrepancies are exported.

SHSS aggregate and BSI MMF portfolio controls. They belong to S12nbfi. No MMF top-up is added to foreign CPIS financial portfolios, which already include MMFs. Government and available central-bank domestic portfolios are retained from financial accounts. Where a central-bank sovereign holding is unpublished, the MFI holder aggregate less SHSS bank and MMF market holdings supplies a flagged residual when the three inputs are available.

3.7 Foreign securities held by nonbank sectors

Domestic holdings are supplemented with foreign securities positions from the IMF's residence-based portfolio survey. Its country-pair totals are more complete than its joint holder–issuer sector cells. We retain published compatible cells and complete the missing sector detail within those totals.

PIP debt-security assets (indicator `P_F3_P_USD`) are classified by holder residence and sector and issuer residence and sector. The financial aggregate S12R includes insurance corporations and pension funds, MMFs and other financial institutions. We use S12R where available and its disjoint components only as alternatives, independently on both axes (IMF, 2017). This avoids adding a parent sector to its children.

Year-end USD positions are converted at the closing ECB exchange rate: 1.0389 USD per euro on 31 December 2024. Quarterly-average FX is not used for this stock conversion. Source observations remain recoverable in their original units.

For a country pair, let T be total-economy holdings, r_h the available holder-sector totals and c_s the issuer-sector totals. The uncovered holder amount $T - \sum_h r_h$ enters an unallocated-holder row. Missing issuer sectors receive the residual to T, with relative weights from their nonresident debt-security liabilities. An observed zero issuer stock therefore does not receive a positive allocation merely to complete the table.

Reported joint holder–issuer cells are fixed when compatible with these margins. The remaining cells are allocated by (4), starting from the independence prior $r_h c_s / T$, after subtracting the fixed cells from each margin. In the present vintage, no proportional shrinking of reported holder margins or adjustment of reported joint cells is required at this stage. Incompatible margins or fixed cells would trigger a separately flagged reconciliation. This is a completion of missing sector detail, not an assertion that PIP publishes a full bilateral sector table.

The completed nonbank portfolios enter the matrix. Bank-holder portfolios serve as allocation priors only: BSI-controlled bank rows replace them, so they are never added as a second bank-security layer. Subsequent issuer reconciliation can adjust retained holdings. Accordingly, PIP country-pair totals are preserved in the intermediate portfolio table, not imposed on the final mixed-source matrix.

3.8 Final checks against issuer stocks and outside positions

The final step checks the holdings assembled from different sources against the securities outstanding. It aligns domestic and foreign valuation margins, resolves overfilled issuer columns and records unidentified holders explicitly. Bank asset totals and issuer totals are both controls; a discrepancy must be reconciled, not hidden in a negative outside position.

Domestic and foreign issuer margins. For each issuer, indexed by u, Eurostat counterpart S2 is the nonresident liability component, not the total amount outstanding. Domestic holdings are recovered from the assets of all resident holders (holder sector S1) against that issuer sector. These distinct margins discipline domestic and foreign portfolios.

Government debt securities use the consolidated nominal debt-security control from the Excessive Deficit Procedure (EDP). Holdings of domestic S13 bonds by S13 itself are removed before

forming the domestic margin, consistently with consolidation. Bank-issued securities use the S122 BSI issuance control; other issuers use their financial-account security stocks. Let M_u^d and M_u^f be domestic and foreign reference stocks and Q_u the issuer control. The common conversion factor is

$$\nu_u = \frac{Q_u}{M_u^d + M_u^f}, \qquad Q_u^d = \nu_u M_u^d, \qquad Q_u^f = \nu_u M_u^f. \tag{5}$$

National-account and CPIS market holdings are converted with this issuer-average factor where needed. Existing ECB face/book inputs retain their source basis. Thus the nominal total is exact, but some individual bank cells remain statistical/book proxies rather than independently verified face amounts. A source-level valuation flag and an accounting-to-promised-payment bridge remain necessary for model use.

If represented holdings exceed a geographic issuer control, imputed positions are reduced first. If that is insufficient, the remaining holdings are reduced proportionally and relabelled reconciled. Raw values and adjustments stay in the register. Unmatched bank security assets return to UNALLOC:ISSUER, preserving the bank instrument control. Thus matching issuer totals can alter individual holdings, and the final matrix distinguishes reconciled cells from source observations.

ROW and unallocated accounts. For issuer u, let $\mathcal{H}_u^d$ and $\mathcal{H}_u^f$ contain represented domestic and foreign holders, respectively, and let W_{vu} be holder v's position in the reporting matrix. With $\mathrm{U_H}$ denoting the unallocated-holder account and ROW the rest of the world, the residuals are

$$W_{\mathrm{U_H},u} = Q_u^d - \sum_{v \in \mathcal{H}_u^d} W_{vu}, \qquad W_{\mathrm{ROW},u} = Q_u^f - \sum_{v \in \mathcal{H}_u^f} W_{vu},$$

after the relevant caps. The first residual is an unidentified domestic holder; the second is outside-sample foreign ownership. Separately, a CPIS holder with known country but unknown sector retains that country in the register. The two types of missing classification are not interpreted as household wealth.

ROW as debtor aggregates observed or allocated bank claims on nonsample euro-area countries and non-euro-area counterparties. The selected nonbank rows cover domestic and sample-country foreign securities; they are not closed to comprehensive world asset portfolios. Their empty ROW-debtor cells do not establish zero nonbank foreign wealth. Likewise, the central-bank node combines measured euro-area central-bank funding/deposit links and available domestic central-bank portfolios in sample-country securities; it is not a complete consolidated Eurosystem balance sheet. Monetary-policy refinancing from ILM is a diagnostic subset, not an extra creditor claim added to the BSI central-bank counterparty position. No capital-key allocation of an aggregate monetary-policy securities portfolio is used in this vintage.

3.9 Register, missing values and reproducibility

Each position records creditor and debtor, instrument, raw/reference and retained amounts, source keys, method, available bounds, inclusion in the matrix and one of five statuses: observed, derived, imputed, reconciled or unallocated. A reference amount can be a portfolio control rather than an observed bilateral position. Source files, metadata, retrieval queries and hashes are retained separately from the constructed exposures.

An unpublished observation is not replaced by an observed zero. Published negative securities positions are retained in an exception register; they receive zero allocation weight/cap in the nonnegative representation while aggregate controls remain binding. A matrix zero means that no positive position is assigned within the retained coverage. It need not mean that the economic exposure is absent.

The reconstruction checks the long register against every matrix cell; loan and deposit positions against their source controls; group removal against gross bounds; bank balance-sheet identities; and all sovereign/security issuer controls. The government columns equal EUR 2,133.0bn (Germany), 2,954.1bn (France), 2,486.9bn (Italy), and 1,452.3bn (Spain). These totals cover debt securities, not all Maastricht debt. All reconciliation records and full-precision matrices accompany the replication files.

4 Bank balance-sheet reconciliation and model interpretation

4.1 Accounting capital and intragroup adjustments

Let A_i be reported statistical assets, K_i capital and reserves, and $D_i^{\text{stat}} = A_i - K_i$ the remaining liabilities. The published BSI identity is

$$A_i = \text{euro-area deposits}_i + \text{issued securities}_i + \text{external liabilities}_i + \text{remaining liabilities}_i + K_i.$$

Instrument and geographical detail are reconciled within these controls. Household bank claims are measured from deposits and securities holdings.

Removing related-office assets $g_i^A = \sum_j G_{ij}$ and funding $g_i^L = \sum_j G_{ji}$ can leave a net cross-border group position. Define

$$n_i^A = \left[g_i^A - g_i^L\right]_+, \qquad n_i^L = \left[g_i^L - g_i^A\right]_+.$$

The following fixed group adjustments preserve capital:

$$A_i^* = A_i - g_i^A + n_i^A, \qquad D_i^* = D_i^{\text{stat}} - g_i^L + n_i^L, \qquad A_i^* - D_i^* = K_i. \tag{6}$$

These net group positions are resources or obligations outside the interbank transmission network. The baseline holds them fixed across sovereign states and banking equilibria.

Statistical capital includes provisions against assets. Let P_i denote those provisions. Showing equity excluding them requires a matched asset adjustment:

$$E_i = K_i - P_i, \qquad A_i^{*,\text{net}} = A_i^* - P_i, \qquad A_i^{*,\text{net}} - D_i^* = E_i. \tag{7}$$

Provisions are not ordinary creditor debt. Nor is either K_i or E_i automatically the model's effective buffer against additional stress losses: existing allowances, valuation and the definition of the shock must be treated consistently. Table 2 reports the accounting benchmarks.

Table 2: Selected balance-sheet controls and reconciled exposures, end-2024. EUR bn unless stated otherwise. Household funding sums measured deposits and reconciled bank securities held by households in the four sample countries; columns identify the debtor banking sector. Bank–bank claims include loans and securities after identified related-office loan removal. Capital components use the public MFI detail; its total equals the corresponding S122 capital control in each sample country. Ratios use adjusted statistical liabilities before fixed outside settlements; they differ from equity divided by clearing debt and are not regulatory leverage ratios.

	Germany	France	Italy	Spain
Reported assets A_i	10,801.2	11,715.2	3,849.0	3,092.4
Related-office assets removed g_i^A	372.4	1,912.1	280.4	221.0
Related-office funding removed g_i^L	430.0	1,993.3	303.3	174.0
Statistical capital K_i	739.8	773.1	376.4	271.4
Asset provisions P_i	53.4	116.1	45.2	25.6
Equity net of provisions E_i	686.4	657.0	331.2	245.7
E_i / D_i^* (percent)	7.08	7.28	10.37	9.28
Household funding of bank sector	3,004.3	1,958.5	1,323.9	1,051.7
Domestic bank–bank claims	1,074.8	1,103.9	112.6	66.4
Domestic bank sovereign bonds	134.4	259.7	356.8	236.6
Reported remaining liabilities	1,914.1	870.9	427.1	205.4

4.2 From accounting positions to repayment exposures

The reporting matrix covers selected bilateral claims, while the source balance sheet also includes government loans, equity holdings, nonfinancial assets and other items. We record those omitted assets and reconcile coverage and valuation differences to the published totals. The remaining net group position is also accounted for separately. Let A_i^{matrix} and D_i^{matrix} be the selected matrix's asset and liability totals, $A_i^{\text{govt loans}}$ separately recorded government loans, and A_i^{other} the other assets and reconciliation adjustments. They satisfy

$$\begin{aligned} A_i^{\text{matrix}} + A_i^{\text{govt loans}} + A_i^{\text{other}} + n_i^A \\ = D_i^{\text{matrix}} + n_i^L + K_i. \end{aligned} \tag{8}$$

Here A_i^{other} is the amount needed to reconcile selected assets with the reported asset total after government loans and related-office adjustments. It combines assets outside the selected claims with coverage and valuation differences; the sources do not identify all of its bilateral counterparties. It is treated as part of reference repayment resources, not as independently measured productive wealth. Model equity cannot be inferred by subtracting the selected column sum from the selected row sum alone.

4.3 Clearing debt and fixed settlements

Write $\mathbf{B} = (B_{ij})$ for the four-country bank-claim matrix and $\mathbf{N} = (N_{ig})$ for bank holdings of sample-country sovereign bonds, with creditor banks in rows. Let O_i collect net group funding, central-bank funding, unallocated remaining funding and outside funding whose repayment rank is not identified. The baseline treats these as fixed settlements, with a matched resource deduction:

$$D_i = D_i^* - O_i, \qquad b_i = D_i + E_i - \sum_j B_{ij} - \sum_g N_{ig}.$$

Thus B_{ij} is bank block i's claim on bank block j, and N_{ig} its sovereign-bond holdings in country g. The common bank recovery pool consists of deposits and issued securities, including creditors

outside the identified four-country block. The repayment estate includes full accounting equity $E_i = K_i - P_i$. The reconciled reference amount b_i includes performing nonbank receipts and outside assets, government loans outside the bond shock, and coverage/valuation adjustments, after settling O_i. It is not separately measured productive wealth. The matched deduction ensures that settling these obligations leaves accounting equity unaffected.

The resulting clearing debts are EUR 7,453.6bn (Germany), 7,987.6bn (France), 2,697.1bn (Italy), and 2,415.9bn (Spain). Fixed outside obligations are approximately EUR 2,235bn, 1,042bn, 495bn and 231bn. The equity-to-clearing-debt ratios are 9.21%, 8.23%, 12.28% and 10.17%; they differ from the statistical-liability ratios in Table 2 because the repayment pools differ.

Remaining liabilities include large derivative and settlement positions. Available MFI derivative-liability benchmarks are EUR 1,505.1bn in Germany, 246.5bn in Italy and 109.2bn in Spain, with corresponding French detail unpublished. These are not exact S122 replacements and do not identify legal close-out rank. Treating the corresponding obligations as fixed settlements is a transparent modelling convention, not a claim that the data identify a common recovery class.

4.4 Scenario inputs and equilibrium outcomes

Exposures, accounting equity, headroom and the default-cost coefficient are inputs to clearing. Sovereign default states, sovereign recoveries and the associated private-loan impairments are specified by the scenario. Bank credit events and repayments are solved jointly; NBFI repayment is endogenous before the full-repayment certificate is applied. The diagnostic of defaults under full interbank repayment evaluates the scenario assuming that every interbank claim is paid in full; NBFIs repay at par under the certificate. Equilibrium feedback then reflects lost interbank receipts, including claims among distinct domestic institutions. This comparison separates direct losses from repayment amplification without imposing a sequential settlement process. Under a guarantee, transfers and repayments are solved jointly with the policy rule and any capacity limit held fixed.

4.5 Public prudential headroom and the distress mechanism

The baseline separates headroom H_i from accounting equity E_i. Public end-2024 Pillar 3 EU KM1 disclosures supply capital and requirement ratios for 56 significant institutions in the four countries. Here j indexes individual institutions. Write $KM1(k)_j$ for row k of the EU KM1 key-metrics disclosure, with ratios in decimal form; RWA_j for risk-weighted assets; $T1_j$ for Tier 1 capital; LE_j for the leverage exposure measure; and OCR_j^{LR} for the overall leverage requirement. Each bank's capacity against the risk-weighted and leverage stacks is

$$h_j^{RW} = [KM1(12)_j - KM1(11)_j]RWA_j, \qquad h_j^{LR} = T1_j - OCR_j^{LR} LE_j,$$

and $h_j^{OCR} = \min\{h_j^{RW}, h_j^{LR}\}$. Writing CET1 for Common Equity Tier 1 capital, row 12 already allows for CET1 used to satisfy Tier 1 and total-capital requirements; row 11 is the combined buffer requirement. We do not subtract the total OCR percentage directly from the CET1 ratio or add headrooms across constraints.

Let $CET1_j$ be institution j's CET1 stock, $CET1_c^{SBS}$ and E_c^{SBS} the country CET1-capital and accounting-equity totals from ECB Supervisory Banking Statistics (SBS), and $E_c^{resident}$ resident accounting equity. The country transfer is

$$H_c = \text{median}_{j \in c}(h_j^{OCR}/CET1_j)(CET1_c^{SBS}/E_c^{SBS})E_c^{resident}. \tag{9}$$

This yields EUR 196.8bn, 208.3bn, 90.6bn and 61.5bn for Germany, France, Italy and Spain. Within-bank ratios cancel native currencies. The public sources are the ECB's selected Pillar 3 disclosures and supervisory statistics (ECB, 2025a). The country proxy uses an unweighted median across banks. The disclosures concern the highest supervisory consolidation level; resident offices, foreign

branches and smaller banks have different coverage. Publication cutoffs also differ: 31 October 2025 for the Pillar 3 compilation and 2 March 2025 for SBS; Germany has 23 versus 24 institutions, respectively. Equation (9) is therefore a labelled perimeter-transfer proxy. The exposure matrix itself retains the residence perimeter.

Requirements, denominators and valuation bridges remain fixed. Losses hit available capital one-for-one, without tax offsets, changing risk weights, new recapitalization, guidance, management targets or an imputed adjustment for minimum requirements for own funds and eligible liabilities (MREL). A breach of buffers does not legally imply default: the regulatory consequences include distribution restrictions, while minimum-capital breaches are distinct (ECB and ESRB, 2026). Market stigma and parallel constraints can nevertheless make buffer use difficult (Magi et al., 2023). We use the published headroom as a reduced-form trigger for costly funding distress in the disorderly sovereign-crisis scenario, conditional on no timely stabilizing response.

Let $\mathcal{L}_{ig}$ denote the additional private-loan loss of bank block i when the sovereign of borrower country g defaults. The next subsection defines this scenario loss from the underlying loan stocks. Normalize contractual bank repayment to one. Let R_j be bank j's repayment rate, $\mathbf{R} = (R_j)$ its vector, $R_g^G(s)$ sovereign g's scenario repayment, and $\delta_g^G(s)$ its default indicator. Write $\lambda_i = \lambda_i^B \in (0, 1]$ for the fraction of the estate retained after distress costs, and V_i for bank i's pre-cost resources (the main paper's A_i). Define

$$V_i(s, \mathbf{R}) = b_i + \sum_g N_{ig} R_g^G(s) - \sum_g \mathcal{L}_{ig} \delta_g^G(s) + \sum_j B_{ij} R_j, \qquad T_i = \min\{D_i + E_i - H_i, D_i/\lambda_i\}.$$

The only bank credit event is

$$R_i = \begin{cases} 1, & V_i \geq T_i, \\ \lambda_i V_i / D_i, & V_i < T_i. \end{cases}$$

We set $\lambda_i = .90$, following the 10% remaining-asset cost benchmark used in the systemic bank-stress model of Aikman and coauthors (Aikman et al., 2009). Its economic interpretation is realization and franchise disruption, rather than administrative expenses alone. The cost is a modelling benchmark for a disorderly crisis. Required capital is not removed from V_i; all remaining accounting equity absorbs losses and costs first. Partial repayment requires both losses $D_i + E_i - V_i > H_i$ and post-cost resources $\lambda_i V_i < D_i$. A headroom breach alone is not sufficient for a creditor loss. The D_i/λ_i cutoff prevents credit-event repayments above face value, and full repayment applies at equality. When $\lambda_i = 1$, the rule reduces to ordinary capped pro-rata clearing, irrespective of $H_i \leq E_i$.

4.6 Private-credit shock and timing

The private-credit shock follows the borrower's country. Let i index the creditor's national bank block and g the borrower's country, each among Germany, France, Italy and Spain. The 4×4 matrix $\mathbf{F}$ records bank loans to nonfinancial firms: F_{ig} is the amount lent by banks in country i to firms in country g. The corresponding matrix $\mathbf{Q}$ records loans to households in country g by banks in country i. Thus rows identify the creditor's country and columns the borrower's country. Both matrices are loan-only extracts from the exposure register, excluding securities contained in the broader sector exposures.

When sovereign g defaults, bank block i incurs the additional private-credit loss

$$\mathcal{L}_{ig} = .018 F_{ig} + .017 Q_{ig}.$$

For example, a French sovereign default generates private-credit losses for German banks on their loans to French firms and households, even if Germany's sovereign does not default. In state s, total additional private-credit losses of bank i are $\sum_g \mathcal{L}_{ig} \delta_g^G(s)$, where $\delta_g^G(s)$ is one if sovereign g

defaults and zero otherwise. These losses are separate from losses on sovereign bonds and from lost interbank receipts.

The rates are rounded exposure-weighted adverse-minus-baseline impairment proxies from the ECB's 2025 stress test, Chart 3.8 (ECB, 2025b). Published secured/unsecured portfolio shares and adverse rates are combined with digitized baseline bars, with the extraction coordinates and rounding preserved. The unrounded pooled incremental rates are approximately 1.81% for firms and 1.65% for households. Corporate and retail regulatory portfolios are proxies for national-accounts borrower sectors, and the public data do not identify country-specific causal losses following sovereign default.

These rates capitalize a three-year credit-stress episode at settlement. They are not annual impairments; the sovereign CDS and safe-return conventions remain annual pricing inputs. The impairment is added once in borrower-country sovereign-default states. Opening equity is already net of provisions, and no provision stock is deducted again. Government loans, private debt securities and outside-sample private loan portfolios receive no additional shock. The exercise does not reconstruct dynamic profits, taxes, credit supply or policy responses.

4.7 Comparison with the ECB's ISA application

The ECB's Interconnected System-wide stress test Analytics (ISA) network at end-2024 uses consolidated bank and fund groups and country-level insurance data, combining supervisory and commercial sources (Grassi et al., 2025). It propagates stress through counterparty losses, liquidity adjustment and fire sales. Our public-data resident-sector construction instead isolates repayment feedback at fixed portfolios. These differences in perimeter and channels preclude a direct comparison of aggregate loss magnitudes.

ISA assumes bank default below a 4.5% CET1 ratio and also models liquidity default. We report headroom above that CET1 floor as a sensitivity; the baseline uses public OCR/leverage headroom as a conditional trigger for costly funding distress, retaining full accounting equity for repayment. The 1.8% and 1.7% loan-loss proxies come from the separate ECB bank stress-test report, not the ISA contagion results (ECB, 2025b). Associating those multi-year losses with sovereign default remains our scenario assumption.

4.8 NBFI repayment and household incidence

For NBFI block m in resident S123–S129, let A_m be financial assets, L_m total liabilities, $\mathbf{F}5_m$ equity and investment-fund-share liabilities, and $BF90_m = A_m - L_m$ net financial worth. The public loss-absorption proxy is financial assets less non-equity liabilities, $E_m^M = A_m - (L_m - \mathbf{F}5_m) = BF90_m + \mathbf{F}5_m$. The common debt pool is F2+F3+F4. Other obligations are fixed outside settlements with matching resource treatment. End-2024 buffers are approximately EUR 4,482.7bn, 2,726.6bn, 1,160.0bn and 917.9bn. Fund net asset value (NAV) and other equity claims can absorb losses, but their national pooling and the nonconsolidated perimeter remain strong approximations. A complete equity cross-holding network is not inferred from these totals.

The eight-node financial system includes bank-to-NBFI, NBFI-to-bank and NBFI-to-NBFI claims, with NBFI repayment endogenous and NBFI lambda equal to one. NBFI buffers exceed the maximum loss on all selected endogenous and sovereign claims even if their receipts vanish. They therefore repay fully in every baseline equilibrium. A separate bound using positive lower bounds on bank repayments establishes par repayment at 50% NBFI availability over the reviewed parameter alternatives. Bank equilibrium vectors coincide exactly under full and half NBFI buffers. NBFI contagion is consequently inactive in this calibration.

Household bank holdings comprise measured deposits and reconciled securities. Deposits account for 95.5–99.8% of resident households' claims on the four bank blocks. Gross claim shortfalls

cannot be equated to uninsured household losses; protection and its financing change incidence (EBA, 2026). The main paper reports the within-state gap $\sum_j x_{H_c,j}(\bar{R}_j - R_j)$ and its percentage of observed annual consumption. A one-for-one consumption interpretation additionally holds nonfinancial resources, other receipts and financing fixed and ignores deposit protection. Consumption levels and household preference parameters are not needed for that difference or for bank clearing.

4.9 Gross domestic interbank claims and exact block aggregation

After the identified related-office adjustment, domestic bank claims are EUR 1,074.8bn, 1,103.9bn, 112.6bn and 66.4bn. They are gross claims among distinct institutions. Retaining them in a national block creates a diagonal in $\mathbf{B}$, rather than an actual own claim by a legal bank. The baseline keeps these exposures and interprets each block as institutions sharing a repayment rate.

A constructive aggregation argument makes this interpretation precise. Split block i into $n_i \geq 2$ identical institutions, each with D_i/n_i, E_i/n_i, H_i/n_i, b_i/n_i and proportional external exposures. For different blocks, allocate claim B_{ij} at $B_{ij}/(n_i n_j)$ per creditor/debtor pair. Within block i, give each ordered pair of distinct institutions claim $B_{ii}/[n_i(n_i-1)]$ and set own claims to zero. Under symmetric repayment R_i, each institution receives $V_i(s, \mathbf{R})/n_i$. Its threshold and debt scale by the same factor, so its repayment map is exactly the aggregate map. Every aggregate fixed point thus lifts to a genuine network without own claims.

All 25 baseline equilibria were checked under two such disaggregations, including unequal numbers of institutions across blocks, with residuals below 10^{-10}. This construction establishes symmetric aggregation of the repayment map. It does not assert symmetric exposure patterns in the data or rule out asymmetric equilibria. Cancelling domestic claims and equal liabilities instead changes both the clearing representation and the estate exposed to default costs; that diagnostic removes baseline multiplicity. Gross domestic exposures are therefore a material aggregation assumption.

4.10 Quantitative results and robustness

The baseline has 12 sovereign states with bank defaults under full interbank repayment and 8 states with multiple equilibria, giving 25 banking equilibria in total. The no-sovereign-default state is unique. The preferred-equilibrium floor leaves one equilibrium in each state and pays zero; a common recovery backstop removes multiplicity at 40% coverage on the five-percentage-point grid. The restricted containment rule, which excludes claims on banks whose own sovereign defaults, pays nothing and leaves the baseline set unchanged. Thus the broad guarantees are effective, while this eligibility restriction is not.

The restricted rule's inactivity holds over the entire coverage interval. The monotone comparison preceding (1) places every equilibrium supported by nonnegative transfers above some original equilibrium. Every eligible bank repays at par in every original equilibrium, so it must also repay at par under the restricted rule. No eligible default can generate a transfer; the policy equations therefore reduce to the original equations, with the same fixed points for every coverage fraction $\eta \in [0,1]$.

The recovery backstop caps bank creditors' receipts at face value, which is necessary because prudential default can occur with positive pre-cost equity. Complete enumeration checks default sets and all binding-floor/cap regions. The sufficient-budget proof above continues to apply to the prudential cutoff. A 75-case joint local sensitivity retains multiplicity in 3–10 states; alternative public quartiles, minimum requirements, sovereign recoveries and buffer-use capacities are reported separately.

The replication files provide the exposure register, source metadata, model inputs and equilibrium calculations. Guarantee comparisons preserve the reference exposures; the separate purchase

counterfactual changes only the specified bond and reserve positions. The R driver `main.R` generates the reported results; amounts are expressed in EUR billions.

5 Accounting for the Eurosystem purchase experiment

The financial calculation conditions on exposures and reconciled net resources. Household-saving optimality and a model of firms' portfolio choices are unnecessary for this conditional exercise. An optional household completion is provided in the main-paper appendix. Probability-weighted repayment and transfer statistics are computed after clearing; they do not determine positions.

Transaction and timing. Before the sovereign state is realized, the Eurosystem purchases a specified fraction of the four bank blocks' holdings of selected sovereign issuers, at the model's book value of one. Let x_{uv} denote a claim from creditor node u to debtor node v, let B_i denote bank block i, G_g sovereign g, and E the Eurosystem. Write $\mathbf{P} = (P_{ig})$ for the purchase matrix, with P_{ig} the purchase from bank block i of sovereign g. In the reporting matrix, the transaction is

$$x'_{B_i,G_g} = x_{B_i,G_g} - P_{ig}, \qquad x'_{E,G_g} = x_{E,G_g} + \sum_i P_{ig}, \qquad x'_{B_i,E} = x_{B_i,E} + \sum_g P_{ig}.$$

The first identity is bilateral; the second applies once to each issuer and the third once to each bank. Bank sales create an equal safe reserve claim on the Eurosystem, consistent with the balance-sheet mechanics described in ECB (2021), Box 3. Government debt columns and bank asset totals are unchanged. Eurosystem assets and liabilities both increase by $\sum_{ig} P_{ig}$. The source matrix is preserved; counterfactual matrices are saved separately.

Let $\mathbf{b} = (b_i)$, $\mathbf{D} = (D_i)$ and $\mathbf{E} = (E_i)$ collect reference resources, clearing debts and accounting equity; $\mathbf{1}$ is the vector of ones. In the financial system, $\mathbf{N}' = \mathbf{N} - \mathbf{P}$ and $\mathbf{b}' = \mathbf{b} + \mathbf{P1}$. All bank debt, accounting equity, headroom, distress cutoffs, interbank claims and private-loan impairments retain their baseline values. In particular,

$$\mathbf{b}' + \mathbf{B1} + \mathbf{N}'\mathbf{1} = \mathbf{D} + \mathbf{E}.$$

Risk-weighted capital requirements are not recomputed. The reserve claim repays one. Since the baseline sovereign repayment exceeds one in survival states under the pricing normalization, the exchange can reduce bank receipts in those states; the complete scenario grid retains this effect.

Risk transfer. Holding bank repayments fixed, the aggregate change in bank resources equals the Eurosystem's additional net cash shortfall:

$$\sum_i \Delta A_i(s) = \sum_{ig} P_{ig}[1 - R_g^G(s)].$$

This measure is negative in survival states when sovereign receipts exceed reserve payments. Its expectation under the existing pricing probabilities is zero by construction of R_g^G, not evidence of zero economic or fiscal cost. The reported maximum is a state-contingent cash exposure. The model does not allocate it among national central banks or feed it back to sovereign or household resources. Actual public-sector purchase arrangements distinguish national holdings and loss sharing (ECB, 2023); the aggregate node cannot identify those incidences. The experiment also abstracts from market-price effects, lending responses and purchases from nonbank sellers.

Clearing and checks. For each purchase matrix, all bank default sets are re-enumerated in each of the sixteen sovereign states. The unchanged NBFI buffer bound continues to imply full debt repayment. Independent eight-node bank–NBFI enumeration verifies all states for purchases of half of all issuer holdings, half of French holdings and three quarters of Italian and Spanish holdings. The zero-purchase case reproduces every baseline equilibrium. Accounting checks verify government debt stocks, bank asset totals, full equity and the equality of direct resource gains and Eurosystem cash exposure. Critical floor capacities are recomputed for the post-purchase network using the same proportional rule, with endpoint checks. The selected published capacities are attained minima; no optimization over purchase composition or quantity is claimed.

The R-only driver `main.R` reproduces the experiment, its publication table and validation. Saved results include state and bank outcomes, counterfactual matrices and validation records. The frozen calibration and underlying exposure register are not overwritten.

References

Jackson, M. O. and A. Pernoud (2021). *Systemic Risk in Financial Networks: A Survey*. Annual Review of Economics 13, 171–202; public version, Section 5.2.

Jackson, M. O. and A. Pernoud (2024). *Credit Freezes, Equilibrium Multiplicity, and Optimal Bailouts in Financial Networks*. Review of Financial Studies 37(7), 2017–2062, p. 2022 and Sections 3.1, 3.3 and 6.1–6.2; public May 2023 version, Appendix B.3, Lemma 1.

European Central Bank (2024). *Manual on MFI Balance Sheet Statistics*, February. ECB methodological manual.

European Central Bank (2026). Balance Sheet Items (BSI), Securities Holdings Statistics (SHSS), Internal Liquidity Management (ILM) and Exchange Rates (EXR). Public data snapshots retrieved 21 October.

Eurostat (2026). Quarterly financial accounts, `nasq_10_f_cp`, Annual financial balance sheets, `nasa_10_f_bs`, and Quarterly government debt, `gov_10q_ggdebt`. Archived source extracts in the replication package.

International Monetary Fund (2017). *Coordinated Portfolio Investment Survey Guide*, third edition. IMF compilation guide.

International Monetary Fund (2026). *Portfolio Investment Positions (PIP), formerly CPIS*. Public SDMX data service; annual 2024 debt-security positions retrieved 21 October.

European Banking Authority (2026). Deposit Guarantee Schemes data and methodological explanation. Accessed 21 October.

Aikman, D., P. Alessandri, B. Eklund, P. Gai, S. Kapadia, E. Martin, N. Mora, G. Sterne and M. Willison (2009). *Funding liquidity risk in a quantitative model of systemic stability*. Bank of England Working Paper 372, Section 2.4.4. Public working paper.

European Central Bank (2025a). Selected Pillar 3 information, end-2024, and Supervisory Banking Statistics, Q4 2024. Cached public workbooks and source-cell extraction in the replication package.

European Central Bank (2025b). *2025 stress test of euro area banks: Final results*, August, Chart 3.8. Public report.

Grassi, A., M. Kosiahn, C. Lelli, M. Losa Martín, M. Moers, M. Sydow, M. Vincent and G. Wiersema (2025). *Integrating contagion risk into the 2025 EU-wide stress test: a system-wide analysis with amplification effects between banks and non-banks*. ECB Macroprudential Bulletin 32, November, Sections 2–3. Public article.

Magi, A., G. Leitner, B. Zsamboki and M. Behn (2023). *How ample is macroprudential space? A simple indicator of effectively releasable buffers measuring macroprudential space*. ECB Macroprudential Bulletin 23. Public article.

European Central Bank and European Systemic Risk Board (2026). *Report of the ECB–ESRB workstream on buffer usability*, April, Box 1. Public report.

Tarski, A. (1955). *A lattice-theoretical fixpoint theorem and its applications*. Pacific Journal of Mathematics 5(2), 285–309, Theorem 1.

Roch, F. and H. Uhlig (2018). *The dynamics of sovereign debt crises and bailouts*. Journal of International Economics 114, 1–13, Section 3.

Segal, I. (2003). *Coordination and discrimination in contracting with externalities: divide and conquer?* Journal of Economic Theory 113(2), 147–181, Section 4; author version.

European Central Bank (2021). *The redistribution of central bank reserves borrowed under TLTRO III*, Box 3, prepared by Sebastiaan Pool, in *TLTRO III and bank lending conditions*. Economic Bulletin 6. Public article.

European Central Bank (2023). *FAQ on the public sector purchase programme*, updated 9 August, Questions 1.6 and 3.1.5. Public FAQ.